Spatial Context Causal Adjustment: An Open Diagnostic Workflow for Geographic Observational Studies

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Abstract

Geographic observational studies estimate effects from spatially structured treatments, outcomes, and environments. Standard causal estimators can be applied, but their credibility depends on whether spatial context is a defensible adjustment source rather than an automatic control, and on whether remaining spatial structure is reported rather than hidden.

We introduce Spatial Context Causal Adjustment (SCCA), an open diagnostic workflow for constructing and auditing spatial-context adjustment sets. SCCA is positioned as an audit protocol, not a new identification theorem. It treats coordinates, geometry, terrain, remote-sensing indices, land cover and neighborhood summaries as candidate context variables that must be justified by support, balance and robustness diagnostics. The workflow standardises feature construction and adjustment-set comparison, then reports common support, post-adjustment balance, spatial block bootstrap, threshold placebo tests, residual spatial autocorrelation, spatial-lag sensitivity, graph sensitivity, variable-role audits, and rule-based evidence grades. Rather than treating residual Moran's I as an identifying statistic, we interpret it as a downgrade diagnostic and calibrate the default material threshold against a 2,880-fit controlled design spanning four spatial-process families (simultaneous- and conditional-autoregressive, distance-kernel, and non-stationary) that varies spatial autocorrelation strength, sample size and confounder strength; the same design shows where the diagnostic is powerful and where it is not.

We evaluate SCCA with synthetic stress tests, a Chongqing urban heat case, a county GIS reproducibility check, and a synthetic positive control. In Chongqing, our preferred specification is a *pre-treatment confounder-only* adjustment set (coordinates, building geometry, and terrain) rather than the best-balanced set, because Sentinel-derived surfaces are possible post-treatment mediators. This specification estimates a modest positive high-rise association with summer land surface temperature (ATT = 0.303°C, 95% CI [0.220, 0.383]); adding the Sentinel surfaces (the best-balanced but over-adjusted full specification) attenuates the estimate to 0.244°C, illustrating why balance alone must not select the causal design.

Because the outcome is retrieved on a ∼1 km MODIS grid while treatment is defined at the building level, roughly seven buildings share one outcome pixel; we make this change-of-support explicit by reporting cluster-robust standard errors clustered on the shared pixel (which widen the interval by ∼40%) and a pixel-aggregated estimand at the identified scale (ATT $\approx 0.55^{\circ}$C). The matched residual Moran's $I = 0.112$ ($p = 0.01$) crosses the default calibrated material threshold of 0.10, so the case is graded as bounded support. We calibrate that threshold not on a single generator but across four spatial-process families (SAR, CAR, exponential-kernel, and non-stationary), and report how the residual-Moran/bias relationship — and hence threshold performance — varies across them. The county dataset (Esri training/demo data) is retained as an ArcGIS-free workflow and spatial-diagnostic check rather than a substantive validation case: its adjusted coefficient was 0.181 but residual Moran's I was 0.313 ($p = 0.010$), and the spatial-lag coefficient fell to 0.145. A clean synthetic positive control is graded core support, confirming that the core/bounded distinction is discriminative rather than constant.

SCCA therefore contributes a bounded protocol for auditing evidence in geographic causal analysis. It does not solve unmeasured spatial confounding or spatial interference; it makes spatial-adjustment claims inspectable, portable across GIS environments, and harder to overstate.

Keywords: spatial causal inference; spatial confounding; causal adjustment; evidence grading; modifiable areal unit problem; open reproducibility

1 Introduction

Geographic information science increasingly supports policy questions that are causal rather than merely descriptive: whether urban form changes heat exposure, whether infrastructure changes disease risk, whether land-use interventions change environmental outcomes, or whether social context affects population health. These questions are usually studied from observational spatial data. Treatment is rarely randomized, spatial units are embedded in environmental gradients, and outcomes are measured at spatial resolutions that do not always match the intervention units.

The main challenge is not the absence of causal estimators. Matching, weighting, regression adjustment, difference-in-differences, spatial panel models, and machine-learning estimators are all available to GIS researchers [Stuart, 2010, Angrist and Pischke, 2009, Athey and Imbens, 2019, Hernán and Robins, 2020]. The harder question is that spatial context can determine both exposure and outcome. Elevation, slope, land cover, access, urban morphology, historical settlement, and administrative structure can all confound a geographic effect estimate. The spatial causal inference literature has emphasised that spatial data introduce complex dependence, possible interference, and spatially structured unmeasured confounding [Reich et al., 2020, Gilbert et al., 2021, Hodges and Reich, 2010, Paciorek, 2010, Dupont et al., 2022, Khan and Berrett, 2023]. If context variables are omitted, an estimate may retain spatial confounding.

If they are added indiscriminately, the estimator may lose common support, amplify noise, or adjust for variables that are not valid confounders, including post-treatment variables.

Modern GIS workflows make this problem more important. Remote-sensing indices, terrain variables, land-cover summaries, geometry, and neighborhood descriptors are now easy to extract at scale [Anselin and Rey, 2014, Brunsdon, 2016], and recent work on spatial confounding has shown that simply adding spatial smooths or fixed effects can corrupt the very treatment coefficient an analyst is trying to recover [Hodges and Reich, 2010, Dupont et al., 2022, Tec et al., 2023]. Geographic causal analysis therefore needs a workflow that asks a practical question: which spatial-context variables improve credible adjustment for this study, under visible diagnostics, and where does the evidence stop?

This paper proposes Spatial Context Causal Adjustment (SCCA). SCCA is a method workflow rather than a new estimator or a new identification theorem. It wraps standard estimators with spatial-context feature construction, adjustment-set comparison, common-support checks, post-adjustment balance diagnostics, spatial robustness tests, and an evidence grade. The central idea is deliberately conservative: spatial context is useful for causal adjustment only when it improves a diagnosed adjustment design, and a positive estimate should be downgraded when spatial diagnostics show that important structure remains.

Our contributions are:

1. A formal SCCA workflow for selecting and diagnosing spatial-context adjustment sets in geographic observational studies, positioned as an audit protocol rather than a universal estimator. The workflow selects the preferred causal specification on *causal-validity* grounds (a pre-treatment confounder set) rather than on post-match balance alone, and treats a better-balanced but possibly over-adjusted set as an explicit comparison.

2. A residual-spatial downgrade rule motivated by SAR residual decomposition and calibrated against controlled data-generating processes, with a multi-family calibration (SAR, CAR, exponential-kernel, non-stationary) that reports where the rule transfers and where it does not, rather than claiming that residual Moran's I identifies unmeasured confounding.

3. A reproducible diagnostic protocol covering common support, covariate balance, spatial block bootstrap, threshold placebo definitions, residual spatial autocorrelation, neighborhood sensitivity, variable-role auditing, change-of-support diagnostics, and evidence grading; the default residual-Moran threshold is calibrated at $t_{\text{Moran}} = 0.10$ and reported together with high-specificity and per-family sensitivity checks.

4. Focused evidence from synthetic estimator stress tests, a Chongqing urban heat analysis as the main empirical case, a county social-capital notebook/GIS run used for ArcGIS-free operational reproducibility and spatial-diagnostic boundary checking, and a synthetic positive control that the grade engine certifies as core support (so the core/bounded distinction is shown to be discriminative).

The empirical ordering follows this claim. Chongqing is the main case because author-controlled spatial processing links candidate context variables directly to the adjustment design. The county case is more open and inspectable, but relies on third-party training/demo data; it is therefore used for ArcGIS-free packaging, spatial downgrading, and interface reproducibility rather than for the main causal claim.

The paper makes a bounded claim. SCCA does not eliminate unobserved spatial confounding, solve spatial interference, or prove that richer adjustment is always better. It makes such claims harder to make without diagnostic evidence.

2 Related Work

2.1 Causal inference with spatial data

Potential-outcome causal inference defines treatment effects by comparing outcomes under treatment and control states [Imbens and Rubin, 2015, Hernán and Robins, 2020]. In observational studies, the identifying assumption is conditional exchangeability: after adjustment for measured confounders, treatment assignment is as-if random. Matching and weighting operationalise this assumption by constructing comparable treated and control groups [Rosenbaum and Rubin, 1983, Stuart, 2010]. In spatial studies, this assumption is difficult because geographic location and environment are themselves structured confounders. Directed acyclic graphs (DAGs) are useful here to separate confounders, mediators, colliders and post-treatment descendants [Pearl, 2009, VanderWeele, 2015]; we adopt this framing for the candidate spatial-context variables (Figure 1).

Spatial econometrics and spatial statistics provide tools for autocorrelation, spillover, and spatial dependence [Anselin, 1988, LeSage and Pace, 2009, Anselin and Rey, 2014]. Spatial causal inference has developed several approaches for unmeasured spatial confounding, spatial interference, and continuous treatments under neighborhood exposure [Reich et al., 2020, Gilbert et al., 2021, Giffin et al., 2020, Papadogeorgou et al., 2022, Schnell and Papadogeorgou, 2020]. An important strand of this literature warns that naïvely adding a spatial smooth or fixed effect to absorb residual autocorrelation can inflate variance and bias the treatment coefficient when treatment itself is spatially patterned, a problem known as spatial confounding bias [Hodges and Reich, 2010, Paciorek, 2010, Dupont et al., 2022, Khan and Berrett, 2023]. Recent work has also extended double-machine-learning estimators to settings with spatial dependence, where cross-fitting blocks must be defined geographically rather than at random [Tec et al., 2023]. These tools are essential, but they do not by themselves select valid adjustment sets for routine GIS workflows. A spatial lag term can reduce residual autocorrelation while leaving treatment assignment confounded; a coordinate trend can absorb broad gradients while failing to represent local land-cover context. SCCA therefore treats spatial dependence diagnostics and causal adjustment diagnostics as related but distinct requirements.

2.2 Spatial context as an adjustment source

Geographic adjustment sets commonly include coordinates, administrative fixed effects, distance measures, terrain, land cover, built environment, and remote-sensing indices. These variables can proxy for otherwise unmeasured context, but proxy adjustment requires substantive assumptions about whether the context source is pre-treatment, related to common causes of exposure and outcome, and measured at a compatible spatial scale [Gilbert et al., 2021, VanderWeele, 2015]. They can also create new problems if they are weakly related to treatment assignment, strongly collinear, measured at incompatible scales, or affected by treatment. The choice of spatial scale and aggregation interacts directly with the modifiable areal unit problem (MAUP) and with edge-effect concerns familiar from geographical analysis [Openshaw, 1984, Fotheringham and Wong, 1991, Manley et al., 2006]: a context variable that is meaningful at one resolution can become uninformative or post-treatment at another. The SCCA workflow therefore requires each candidate context source to pass common-support and balance checks before it is used to support a causal statement, and reports graph-specification and threshold sensitivity to expose how scale choices change the conclusion.

2.3 Continuous exposure, spatial interference, and diagnostics

Many GIS applications use continuous or ordered exposures rather than binary treatments. Generalised propensity-score methods extend the propensity-score logic to continuous treatments and dose-response estimation [Hirano and Imbens, 2004, Zhao et al., 2013]. Spatial settings add a further complication because one unit's exposure can affect outcomes in nearby units, violating the simplest no-interference assumption. Spatial interference methods address this directly in specialised designs, ranging from the early Verbitsky-Savitz framework [Verbitsky-Savitz and Raudenbush, 2012] to modern exposure mappings and identification results under partial interference [Aronow and Samii, 2017, Forastiere et al., 2021, Giffin et al., 2020, Papadogeorgou et al., 2022]. SCCA does not claim to identify spillover effects in this formal sense. Instead, it reports neighborhood-exposure and spatial-lag sensitivity outputs as diagnostics for whether a local adjusted estimate is stable after explicit neighborhood terms are introduced.

2.4 Diagnostics and evidence grades

Causal estimates are often reported as single numbers even when the adjustment design is fragile. For geographic studies, this is especially risky because residual spatial structure can indicate remaining confounding or interference. SCCA follows a diagnostic-first stance: effect estimates are interpreted together with post-adjustment standardized mean differences, matched sample counts, common-support attrition, bootstrap stability, placebo checks, residual spatial autocorrelation, neighbor-exposure sensitivity, and graph-specification sensitivity. The final evidence grade is not a statistical significance label; it is a compact description of whether the adjustment design provides core or bounded support under a declared rule table.

2.5 Position relative to adjacent workflows

SCCA is an audit layer that connects causal adjustment diagnostics with GIS-facing spatial diagnostics. It reuses standard matching, weighting, regression, GPS and SLX-style sensitivity models, but adds a shared request schema, alternative context variants, common-support and balance checks, residual-spatial downgrades, and portable GIS outputs. Relative to the spatial-confounding literature it builds on [Hodges and Reich, 2010, Paciorek, 2010, Dupont et al., 2022, Khan and Berrett, 2023, Gilbert et al., 2021, Tec et al., 2023], the increment is not a new estimator or identification result but three operational contributions: (i) selecting the reported specification on a declared causal-role basis (pre-treatment confounders) rather than on fit or balance, with the better-balanced set kept as an explicit over-adjustment comparison; (ii) turning residual spatial autocorrelation from a modelling nuisance into a pre-registered, machine-readable downgrade rule whose material threshold is calibrated and whose power is characterised across spatial-process families; and (iii) carrying these decisions, plus change-of-support diagnostics, into one portable evidence contract across GIS interfaces. The intended contribution is therefore a reproducible protocol for asking whether a spatial-context adjustment set is usable for a bounded causal statement and whether that statement should be downgraded after spatial diagnostics, not a replacement for any single method class or proprietary GIS interface. The need for such an audit layer is consistent with recent GIScience work on reproducibility and reusable computational geography [Nüst and Pebesma, 2021, Konkol and Kray, 2019, Brunsdon, 2016].

3 Spatial Context Causal Adjustment

3.1 Problem formulation

Let $i = 1, \ldots, n$ index geographic units with location or geometry g_i. Each unit has treatment or exposure T_i, outcome Y_i, observed non-spatial covariates X_i, and candidate spatial-context variables C_i. The context vector may contain coordinates, geometric descriptors, terrain, land-cover summaries, remote-sensing bands or indices, and neighborhood summaries. Figure 1 represents these variables as a directed acyclic graph for the Chongqing case: terrain and historical land-use act as classical observed confounders, a latent spatial process U proxies the unmeasured spatial structure SCCA cannot identify by itself, and a candidate context variable such as NDVI is shown explicitly as a potential mediator that SCCA must decide whether to include as an adjustment variable or to keep out of the adjustment set.

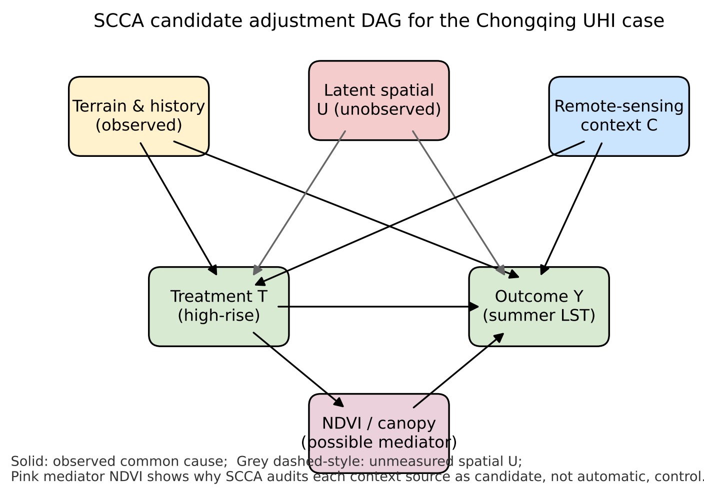

Figure 1: Candidate adjustment DAG for the Chongqing UHI case. Terrain and historical land-use are observed common causes of treatment T (high-rise) and outcome Y (summer LST). A latent spatial process U remains unobserved; SCCA reports residual Moran's I and spatial-lag sensitivity as diagnostics for the resulting bias rather than identifying U. NDVI is shown explicitly as a possible post-treatment mediator: SCCA's variable-role layer requires the analyst to declare whether it is added as confounder or omitted as descendant of T.

For binary treatment, the target estimand in the main case study is the average treatment effect on the treated:

$$\tau_{\mathrm{ATT}} = E[Y_i(1) - Y_i(0) \mid T_i = 1]. \tag{1}$$

For continuous exposures, SCCA estimates exposure-response functions or local slopes, but the same adjustment logic applies.

SCCA does not identify τ by itself. Identification still requires that relevant confounders are measured, treatment has sufficient overlap, and interference is limited or explicitly modeled. SCCA contributes a structured workflow for testing whether a chosen spatial-context adjustment set is plausible enough to support a bounded causal interpretation.

3.2 Identification assumptions and scope

The workflow rests on standard observational-causal assumptions, made explicit for spatial data [Imbens and Rubin, 2015, Pearl, 2009]. Consistency requires that the exposure, outcome, and spatial unit match the intervention being interpreted. Conditional exchangeability requires that observed covariates and candidate spatial context contain the relevant common causes of exposure and outcome; coordinates and remote-sensing variables are only proxies, not guarantees [Gilbert et al., 2021]. Positivity requires usable overlap, so SCCA reports support loss and balance rather than only fitted coefficients. No-interference must be plausible or bounded; neighbor-exposure and spatial-lag outputs are sensitivity checks, not identified spillover estimands. Scale compatibility is also required because exposure, context, and outcome may be measured at different resolutions. The evidence grade is therefore a claim-strength label under these assumptions, not a substitute for them.

3.3 Workflow

Figure 2 summarizes the workflow. The first step builds candidate context features. The second step defines adjustment variants, from minimal baselines to richer context sets. The third step estimates propensity scores, generalized propensity scores, or outcome models depending on the exposure type. The fourth step diagnoses common support and balance. The fifth step runs robustness checks and assigns an evidence grade.

The estimators are deliberately standard. The contribution is the workflow that treats spatial context as a diagnosable adjustment source, carries spatial diagnostics into the claim label, and writes the same evidence contract across GIS interfaces.

3.4 Adjustment variants

SCCA does not impose a single estimator on all exposure types. The reusable core estimates adjusted outcome models and ERFs for continuous or ordered exposures using generalized propensity-score (GPS) style weights [Hirano and Im-

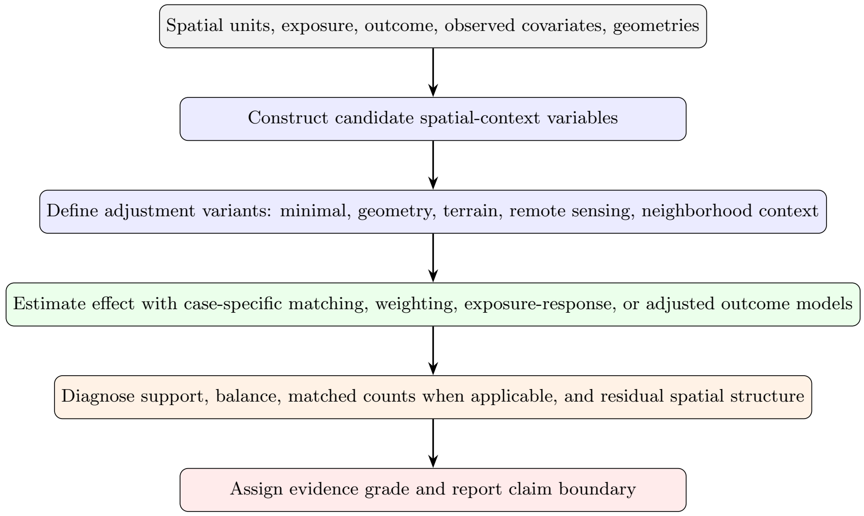

Figure 2: SCCA workflow. Spatial context is treated as a candidate adjustment source and must pass support, balance, and robustness diagnostics before it is used to support a causal claim.

bens, 2004, Zhao et al., 2013]. Case-specific modules can add binary-treatment estimators when the scientific question is naturally treated-control. In the Chongqing module used below, the binary module estimates a propensity score

$$e_i = P(T_i = 1 \mid X_i, C_i) \tag{2}$$

for each adjustment variant. Units outside the overlapping propensity-score range are removed. Treated units are matched to controls under a nearest-neighbor caliper rule, and standardized mean differences (SMDs) are computed before and after matching:

$$\mathrm{SMD}_k = \frac{\bar{Z}_{k,T=1} - \bar{Z}_{k,T=0}}{\sqrt{(s^2_{k,T=1} + s^2_{k,T=0})/2}}, \tag{3}$$

where Z_k is the kth covariate in the adjustment set. A variant is treated as credible only if the maximum post-match absolute SMD is below 0.1 unless a case-specific justification is provided. The estimate is then reported together with matched sample counts and common-support attrition.

For the shared continuous-exposure workflow, SCCA reports baseline adjusted OLS, difference-outcome OLS when a baseline outcome is available, and a generalized propensity-score weighted exposure-response function (ERF). The ERF is fit as a weighted outcome model over an exposure grid. The implementation writes the point estimate, pointwise model-based confidence limits for the fitted response, and an approximate confidence interval for the ERF range effect. These intervals describe uncertainty in the fitted outcome model; they do not replace the spatial bootstrap, placebo, support, and residual-spatial diagnostics used for the evidence grade.

3.5 Robustness diagnostics

SCCA includes four robustness checks when data permit them: spatial block bootstrap, placebo exposure definitions, weaker-outcome or placebo-outcome checks, and residual spatial autocorrelation diagnostics. Together they test whether the estimate is stable under spatial dependence, threshold choices, weaker causal contrasts, and remaining spatial structure.

For spatial datasets with geometries or coordinates, SCCA builds a lightweight neighborhood graph from polygon adjacency or coordinate k-nearest neighbors. The graph supports exposure and residual Moran diagnostics, a neighbor-exposure adjusted estimate, graph-specification sensitivity, and an SLX-style sensitivity model [LeSage and Pace, 2009]:

$$Y_i = \alpha+\beta T_i+\theta(WT)_i+\gamma X_i+\phi C_i+\eta(WX)_i+\lambda(WC)_i+\varepsilon_i, \qquad \varepsilon \sim \mathcal{N}(0,\sigma^2 I), \tag{4}$$

where W is a row-standardised neighborhood matrix. Spatially correlated errors are handled separately by residual Moran's I and spatial block bootstrap, rather than by simultaneously fitting a SAR error model. We interpret β as the local adjusted association and θ as a neighboring-exposure signal. This is a stability diagnostic, not a full interference design.

3.6 Diagnostic interpretation of residual Moran's I

The residual-Moran rule is used as a warning boundary, not as an identification result. To motivate the rule, consider a simple data-generating process

$$Y_i = \tau T_i + X_i^\top \gamma + C_i^\top \phi + \sigma_u U_i + \nu_i, \tag{5}$$

where $U = (I - \rho W)^{-1}\epsilon$ is a stationary SAR(ρ) field with $|\rho| < 1$, $\epsilon \sim \mathcal{N}(0, I)$, and ν_i is i.i.d. noise independent of U. If SCCA adjusts for (X, C) but not for U, the fitted residual can be written heuristically as

$$r \approx M\nu + \sigma_u MU, \tag{6}$$

where M is the projection that removes the fitted adjustment design. This decomposition explains why residual Moran's I tends to increase when an omitted spatially structured component remains after adjustment. It does not identify ρ, σ_u, or the treatment bias, and it does not prove that a non-significant Moran statistic rules out spatial confounding.

SCCA therefore treats residual Moran's I as an empirical downgrade diagnostic. The rule asks a narrower question: after the declared adjustment set has been fitted, is the remaining spatial structure large enough to make a stronger causal statement unsafe under the manuscript's rule table? Section 4.8 calibrates that materiality threshold in a controlled SAR-confounded DGP, and

Section 4.9 reports how the case-study grades change when the residual-Moran threshold is varied. This calibration makes the warning rule auditable, but it remains a diagnostic boundary rather than a theorem about identification.

3.7 Evidence grades and threshold calibration

SCCA assigns evidence grades using a machine-readable rule file. *Core support* requires a strong credibility decision, robust support, and no material spatial downgrade. *Bounded support* marks useful evidence with explicit support, scale, balance, robustness, residual-spatial, or reproducibility limits. Manuscript claims use only core and bounded evidence rows.

The manuscript uses the 2026-06-30 evidence-grade rule set. A case is downgraded from core to bounded support when credibility is moderate or weak, exposure-balance correlation exceeds 0.50, exposure boundary mass exceeds 0.25, robustness is bounded or fragile, $|I_{\text{residual}}| \geq t_{\text{Moran}}$ with permutation $p \leq 0.05$, the neighbor-exposure coefficient remains significant at $p \leq 0.05$, spatial adjustment changes the main coefficient by at least 25%, or graph-sensitivity checks do not preserve the effect sign. The default material residual-Moran threshold is $t_{\text{Moran}} = 0.10$ in this rule set. The full table is exported as `07_results/scca_evidence_grade_rules.json` and `07_results/scca_evidence_grade_rules.md`.

These thresholds are audit rules, not universal validity constants. The SMD threshold 0.10 follows matching practice [Stuart, 2010, Austin, 2009]; the 25% coefficient-change, 0.25 boundary-mass and 0.50 balance-correlation rules are manuscript defaults. The residual-Moran threshold is explicitly calibrated (Section 4.8) by sweeping a controlled DGP across ρ, n and σ_u/σ_ν and comparing true-positive and false-positive rates for materially biased SCCA estimates. The default 0.10 is conservative but more sensitive than the high-specificity 0.20 threshold retained as a sensitivity check.

A downgrade does not prove bias, and its absence does not prove identification. It only states whether the estimate crossed a declared warning boundary. To avoid hiding sub-threshold spatial structure, the implementation records `significant_residual_moran_below_material_threshold` when residual Moran's I is statistically significant but below the material threshold. Threshold overrides are written to `07_results/scca_grade_threshold_sensitivity.csv`, so grades and boundary conditions come from the same rules rather than post-hoc narrative judgement.

3.8 Implementation surfaces

SCCA is implemented as a shared Python core with thin command-line, notebook, ArcGIS Pro, and QGIS Processing adapters. The reusable `AnalysisRequest` object records input data, exposure and outcome fields, covariates, context columns, coordinates, bootstrap groups, placebo settings, trimming thresholds, target outcomes, and explicit *variable-role* declarations. Each run writes CSV, JSON, and Markdown diagnostics and, for spatial inputs, GeoPackage, GeoJSON, Shapefile, static/interactive maps, and QGIS `.qml` styles. This architecture keeps ArcGIS optional and prevents interface-specific implementations from drifting.

The binary Chongqing module separately writes ablation, balance, and matched-count tables used for the case-study ATT, SMD, and retained-sample summaries.

3.9 Missing data, scale, and uncertainty

Three implementation choices remain deliberately conservative and are recorded in the manuscript rather than buried in software. Missing context values are handled by within-variant median imputation with an explicit missingness indicator; the missingness indicator is added to X so that any covariate-pattern shift due to imputation is captured by the post-match SMD. Analyses with spatially structured missingness should additionally rerun the workflow under complete-case removal as a sensitivity check. Scale and MAUP [Openshaw, 1984, Fotheringham and Wong, 1991, Manley et al., 2006] interact with SCCA at three points: the spatial unit on which T is defined, the resolution at which C is extracted, and the neighborhood graph used for spatial diagnostics. When the outcome is retrieved at a coarser resolution than the treatment unit — as in the Chongqing case, where a building-level treatment meets a $\sim$1 km LST outcome — SCCA reports an explicit *change-of-support* panel (Section 4): the building-level estimate, the same estimate with standard errors clustered on the shared outcome pixel, and a pixel-aggregated estimand at the outcome resolution. SCCA also reports graph-sensitivity across $k \in \{4, 6, 8, 12\}$ for coordinate-kNN graphs and threshold-placebo sensitivity for binary exposures, but a single SCCA run does not solve scale mismatch by itself. Uncertainty intervals come from four distinct sources, and we report which source is used in each table: (i) cluster-robust HC3 standard errors for adjusted OLS and SLX; (ii) standard errors clustered on the shared outcome pixel for the change-of-support panel; (iii) the empirical 2.5%–97.5% quantile of the spatial block bootstrap ($B = 200$ for Chongqing, $B = 50$ for the county case, using contiguous coordinate blocks whose side is set to roughly twice the empirical range of the residual variogram); and (iv) Abadie–Imbens robust matching standard errors for the binary ATT [Abadie and Imbens, 2006].

4 Experiments

4.1 Dataset roles and case selection

The experiments assign different evidentiary roles to the available datasets. Chongqing is the main real-data SCCA case because the scientific question, building-level high-rise treatment, geometry/terrain/remote-sensing context, and summer LST outcome are tightly coupled. It directly tests whether richer spatial context can be treated as a diagnosed adjustment source without overstating the association.

The county social-capital data are used differently. They are familiar to GIS users and allow comparison with a commercial GIS causal-inference example, but they are third-party training/demo data governed by source terms. They therefore test ArcGIS-free rerun counts, output contracts, residual spatial diagnostics, and downgrading behavior rather than serving as primary empirical validation.

The reproducibility scope also differs by dataset (Table 1). The public repository supports structural Chongqing reruns and exact reruns of redistributable examples, but it does not redistribute raw building-level Chongqing inputs.

Table 1: Reproducibility scope of the experimental cases.

Case	Public rerun level	Restricted or source-governed inputs	Manuscript use
Synthetic benchmark audit	Exact public rerun from committed generators and scripts	None beyond software dependencies	Stress-test estimator behavior under known targets
Chongqing urban heat	Structural rerun from public scripts, manifests, and generated summaries	Building footprints, precise coordinates, floor attributes, and derived local environmental variables require approved local access	Main real-data SCCA ablation and claim-boundary example
County social capital GIS run	Exact rerun where users have the licensed third-party example dataset; public outputs and scripts document the contract	Esri training/demo data and source terms govern redistribution and use	ArcGIS-free GIS workflow, cross-interface output, and spatial-diagnostic downgrade check

4.2 Evidence synthesis design

The experiments addressed four questions: estimator behavior under controlled generators; balance, effect estimation, and change-of-support in the Chongqing urban heat case; GIS/notebook reproducibility with spatial-diagnostic boundary checking in the county social-capital run; and whether the evidence-grade engine can certify a clean design as core support (a positive control, `07_results/core_support_positive_control.json`).

Table 2 summarizes the resulting evidence matrix. The source table is generated by the synthesis script and stored in `07_results/scca_evidence_synthesis.csv`; the same script writes residual-spatial threshold sensitivity to `07_results/scca_grade_threshold_sensitivity.csv`.

4.3 Synthetic benchmark audit

The controlled benchmark used six estimator families, including matching, DiD, ERF, causal forests, spatial Granger tests, and GCCM. These rows are stress tests, not empirical claims. The audit produced 48 summary rows: DiD and causal forests recovered their baseline targets, ERF remained useful but biased in some settings, and direction-recovery or severe-stress settings were fragile. The benchmark therefore supports SCCA as a diagnostic wrapper around esti-

Table 2: SCCA evidence synthesis. Grades describe claim strength, not statistical significance alone.

Case	Data type	Key diagnostic	Grade	Manuscript use
Synthetic benchmark audit	Controlled generators	48 benchmark rows; 13 robust and 29 fragile rows	Bounded support	Estimator stress test, not real-world validation
Chongqing urban heat	Real building and remote-sensing data	Pre-treatment ATT = 0.303°C (over-adjusted full = 0.244); pixel-aggregated ≈0.55; residual Moran's I = 0.112	Bounded support	Main real-data SCCA case with change-of-support and residual-spatial caution
County social capital spatial notebook	GIS/notebook reproducibility case	residual Moran's I = 0.313; spatial-lag coef. = 0.145; ArcGIS-trimmed ERF range = 6.85 years	Bounded support	Spatial diagnostics and cross-interface output check
Synthetic core-support control	Clean controlled DGP (all confounders measured)	$\widehat{\text{ATT}}$ = 0.505 (true 0.5); residual Moran's $I \approx 0$ (n.s.); no downgrade rule fires	Core support	Positive control showing the core/bounded distinction is discriminative

mators, not as a guarantee that every estimator works in every spatial setting. Full rows are reported in the generated benchmark audit outputs.

4.4 Chongqing urban heat case

The main case asked whether Chongqing high-rise buildings were associated with local summer land surface temperature after spatial-context adjustment. The analysis used a stratified 2021 sample of 5,000 buildings. Treatment was defined as at least 10 floors; the matched sample retained 1,621 treated buildings (10–53 floors; median 14, interquartile range 11–19), so the ATT mainly describes mid-rise to upper-mid-rise contrasts rather than ultra-tall outliers. Summer LST was retrieved from the MODIS MOD11A2 8-day product (June–August 2021) on its native ∼1 km grid and sampled at each building centroid. Candidate context variables included coordinates, footprint area, Sentinel-2 bands, NDVI, NDBI, MNDWI, NDWI, 30 m SRTM elevation, and slope. The ∼1 km outcome resolution is coarser than the building treatment unit, a change-of-support issue we quantify explicitly below rather than treat as a footnote.

Preferred causal specification. We report the *pre-treatment* confounder-only adjustment set (coordinates, building geometry, and terrain) as the preferred causal specification, chosen on causal-validity grounds rather than on minimum post-match SMD. This set uses only context that is plausibly fixed before high-rise construction. It estimated ATT = 0.303°C with 95% CI $[0.220, 0.383]$ and maximum post-match SMD = 0.104 (Table 4). By contrast, the best-*balanced* specification – the full remote-sensing set that adds Sentinel bands and indices – gave a smaller ATT = 0.244°C (max post-match SMD = 0.061). The two differ by about 19%, and the attenuation is expected if the added Sentinel surfaces partly encode the treatment's own thermal footprint (over-adjustment for a mediator). This is precisely why balance quality must not, by itself, select the causal design: the better-balanced set is the less causally defensible one. The CIs use Abadie–Imbens robust matching standard errors

[Abadie and Imbens, 2006]; a $B = 200$ spatial block bootstrap gives closely overlapping intervals and sign-stability of 1.000. Threshold placebos at 8, 10, and 12 floors are consistent with dose-response rather than a sharp threshold effect (Figure 4).

Change of support. Because the outcome is retrieved at $\sim$1 km while treatment is defined at the building level, many buildings share one outcome pixel. In the analysis sample the 5,000 buildings map to about 1,400 distinct outcome pixels (roughly 3.6 buildings per pixel in the modelled subset; up to $\sim$7 in the raw sample), and within a shared pixel the buildings carry an identical outcome value. A building-level standard error that ignores this shared measurement overstates precision. Table 3 therefore reports three estimands for the pre-treatment set: (i) a naive building-level adjusted OLS, (ii) the same point estimate with standard errors clustered on the shared outcome pixel, and (iii) a pixel-aggregated model that collapses to one row per pixel and regresses pixel-mean LST on pixel treatment share. Clustering widens the interval by roughly 40% (SE 0.038 $\rightarrow$ 0.054), and the pixel-aggregated estimand is materially larger (ATT $\approx$ 0.55°C). We interpret the pixel-aggregated figure as the effect that is actually identified at the outcome's resolution and the building-level figure as a finer-scale approximation whose precision must be discounted; the gap is a property of the data, not a defect the workflow can remove.

Table 3: Chongqing change-of-support diagnostics for the pre-treatment adjustment set (source: `07_results/chongqing_change_of_support.csv`). The building-level point estimate is unchanged by clustering, but the cluster-robust SE (clustered on the shared $\sim$1 km outcome pixel) is $\sim$40% wider, and the pixel-aggregated estimand differs materially.

Estimand	ATT (°C)	SE	95% CI	Units
Building, naive iid SE	0.268	0.038	[0.193, 0.343]	5,000 buildings
Building, cluster-robust SE	0.268	0.054	[0.162, 0.374]	1,400 pixel clusters
Pixel-aggregated	0.546	0.096	[0.359, 0.734]	1,400 pixels

Residual Moran's I remained statistically significant for the pre-treatment specification ($I = 0.112$, permutation $p = 0.01$) and for the over-adjusted full-context set ($I = 0.102$, $p = 0.01$). We report this honestly: the matched residual is not free of spatial structure, and the pre-treatment set leaves slightly more residual structure than the over-adjusted set, consistent with its adjusting for fewer (possibly post-treatment) surfaces. Under the 2026-06-30 evidence-grade rule set, the default material-spatial-caution threshold is $t_{\text{Moran}} = 0.10$, so the preferred specification's residual triggers `material_residual_moran` and the Chongqing case is graded as bounded support. Figure 5 shows the empirical Moran scatter of the matched pre-treatment residual. Under a high-specificity sensitivity threshold of 0.20, the same residual would be recorded as statistically significant residual spatial structure below the material threshold rather than

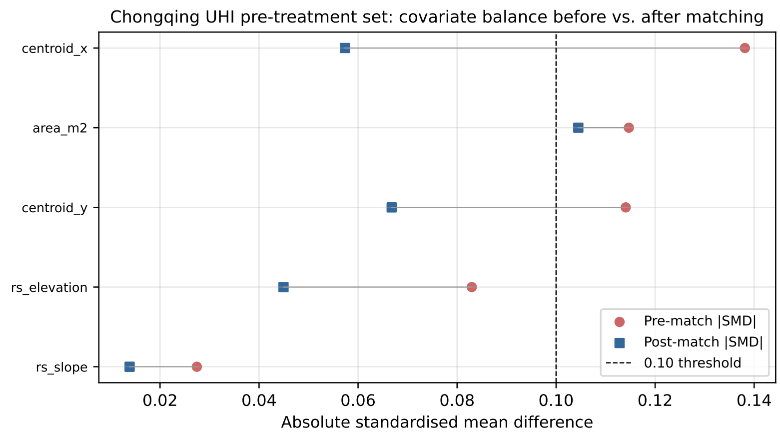

Figure 3: Chongqing UHI covariate balance before and after caliper matching (Love plot) for the pre-treatment set. The 0.10 SMD threshold is shown for reference.

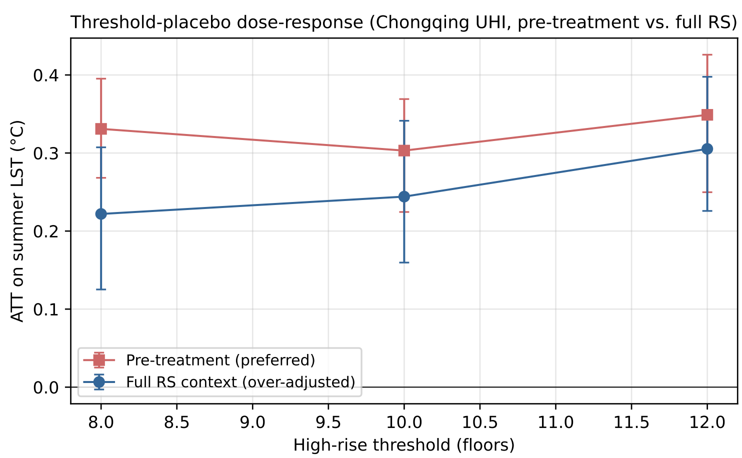

Figure 4: Chongqing threshold-placebo dose-response. Vertical bars are matching-based 95% CIs. The monotone increase from 8 to 12 floors is consistent with a dose-response interpretation; all estimates remain small relative to the thermal-retrieval and change-of-support uncertainties that accompany $\sim$1 km LST extraction.

Table 4: Chongqing UHI adjustment variants. *pre_treatment* (coordinates + geometry + terrain) is the preferred causal specification; *full RS context* adds Sentinel surfaces and is the best-balanced but over-adjusted comparison. The maximum pre-match SMD reaches 0.703 for the full RS context, reflecting strong baseline imbalance in NDBI and Sentinel-2 bands. Balance pass means maximum post-match SMD < 0.10.

Variant	ATT (°C)	CI lower	CI upper	Max pre SMD	Max post SMD	Role
Raw difference	0.238	0.163	0.307	–	–	unadjusted
Coordinates only	0.267	0.178	0.346	0.138	0.080	pre-treatment
Geometry	0.252	0.180	0.333	0.138	0.125	pre-treatment
Terrain	0.303	0.229	0.367	0.138	0.104	pre-treatment
Pre-treatment (preferred)	**0.303**	**0.220**	**0.383**	**0.138**	**0.104**	**confounder-only**
Sentinel indices	0.157	0.051	0.239	0.703	0.083	possible mediator
Sentinel bands	0.121	0.034	0.210	0.463	0.051	possible mediator
Full RS context	0.244	0.126	0.345	0.703	0.061	over-adjusted
PCA context	0.231	0.140	0.324	0.560	0.065	over-adjusted

as a material downgrade. The appropriate scientific claim is therefore a modest measured-context association with explicit residual-spatial caution, change-of-support between the building treatment and the ~1 km LST outcome, and possible interference between adjacent high-rise clusters.

The variable-role audit in Table 5 makes this explicit: remote-sensing features are not all admissible confounders. Coordinates and terrain are low-risk context variables, building geometry is a medium-risk morphology proxy, and Sentinel-derived indices/bands are ambiguous proxies or possible mediators because they may reflect post-construction surfaces. The preferred pre-treatment set therefore trades slightly weaker post-match balance for a more defensible causal interpretation, while the full-RS set is reported as an over-adjustment comparison. The source table is written to `07_results/chongqing_variable_role_audit.csv`.

Table 5: Chongqing variable-role audit for candidate spatial-context sources. Risk describes possible post-treatment or mediator contamination, not statistical balance.

Context group	Causal role	Risk	Variant	ATT (°C)	Max post SMD	Balance pass
Coordinates	Spatial proxy confounder	Low	coordinates only	0.267	0.080	Yes
Building geometry	Proxy confounder or pre-treatment morphology	Medium	geometry	0.252	0.125	No
Terrain	Pre-treatment confounder	Low	terrain	0.303	0.104	No
Pre-treatment (preferred)	Confounder-only adjustment set	Low	pre_treatment	0.303	0.104	No
Sentinel indices	Ambiguous proxy or mediator	High	sentinel indices	0.157	0.083	Yes
Sentinel bands	Ambiguous proxy or mediator	Medium	sentinel bands	0.121	0.051	Yes
Full RS context	Over-adjusted (possible mediators)	Medium	full RS context	0.244	0.061	Yes

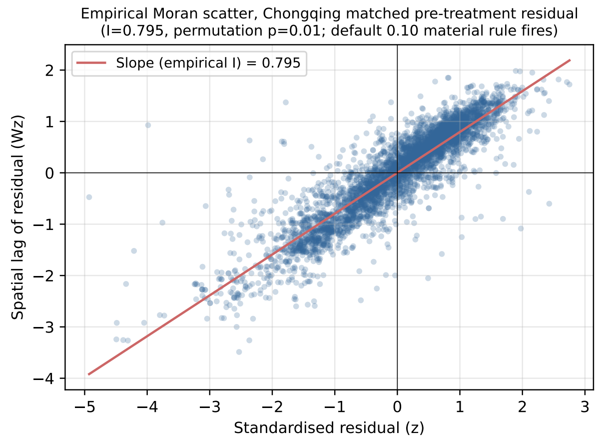

Figure 5: Empirical Moran scatter for the Chongqing matched pre-treatment residual (standardised residual on the x-axis, its spatial lag on the y-axis). The slope is the empirical Moran's $I = 0.112$, statistically significant under 99 permutations ($p = 0.01$); under the default 0.10 rule it triggers a material residual-spatial caution, while under a high-specificity 0.20 sensitivity threshold it would be recorded as significant-below-material.

4.5 County ArcGIS-free GIS and spatial-diagnostic validation

The county run is narrower than the Chongqing case. It tests whether the shared SCCA core can reproduce a familiar GIS-facing county analysis in an open notebook workflow, write portable GIS outputs, and show whether spatial diagnostics change the interpretation. The exposure is a county-level social-association measure from the 2019 County Health Rankings / Esri demo dataset, the outcome is average age at death, and the unit is the contiguous U.S. county.

The county notebook run adds the spatial-diagnostic layer used in the evidence table. After 1%–99% exposure trimming, 3,044 of 3,108 county records remained, matching the count used in an ArcGIS Causal Inference example for the same dataset. Baseline adjusted OLS estimated 0.181, while the spatial-lag sensitivity estimate was 0.145. Residual Moran's I remained 0.313 ($p = 0.010$), and the neighbor-exposure term remained significant ($p < 0.001$). The rule table therefore assigns bounded support through `material_residual_moran` and `significant_neighbor_exposure`: the positive association is stable enough for an operational check, but not for a strong causal claim.

The ArcGIS-compatible county run is therefore a practical reproducibility check, not independent identification evidence or a claim to reproduce proprietary ArcGIS internals. It supports the narrower claim that an ArcGIS-free workflow can reproduce and audit a GIS-facing analysis while retaining explicit evidence boundaries.

4.6 Core-support positive control

Because all three preceding cases are graded bounded support, an engine that always returned bounded support would be observationally indistinguishable from ours on this evidence. We therefore include a positive control (`scripts/core_support_positive_control.py`, output `07_results/core_support_positive_control.json`). It generates a clean observational DGP with $n = 3{,}000$ in which all confounders are measured, treatment overlap is excellent, and no latent spatial process is injected, then runs the SCCA adjusted estimator and the same diagnostics used for the real cases. The control recovers the true effect (ATT $= 0.505$ vs. true 0.5; relative bias $< 1\%$), the post-adjustment covariate SMD is below 0.10, residual Moran's $I \approx 0$ with a non-significant permutation p, and the neighbor-exposure term is non-significant. Under the same rule table, the grade engine returns *core support*: no downgrade rule fires. This confirms the core/bounded distinction is discriminative — the empirical cases are downgraded because of real diagnostics, not because the engine is incapable of issuing a core grade.

4.7 Baseline-to-SCCA comparison

The direct comparisons are stored in `07_results/scca_method_comparison.csv`. In Chongqing, the raw difference was 0.238°C and the preferred pretreatment SCCA estimate was 0.303°C, while the over-adjusted full set gave 0.244°C; SCCA's contribution is not a single coefficient but the balance, variable-role, change-of-support and residual-spatial diagnostics that identify which coefficient is defensible and downgrade the claim under the default rule. In the county case, adding the spatial layer reduced the coefficient from 0.181 to 0.145 and downgraded the evidence because residual Moran's I and neighbor exposure remained significant.

4.8 Threshold calibration for the residual-Moran rule

A central concern for rule-based audit workflows is threshold origin, and a second is whether a threshold calibrated on one spatial process transfers to others. We therefore calibrated the material residual-Moran threshold t_{Moran} with `scripts/threshold_calibration.py` across *four* spatial-process families for the latent confounder: a simultaneous autoregressive (SAR) field, a conditional autoregressive (CAR) field, a distance-based exponential/Matérn-$\frac{1}{2}$ kernel field, and a non-stationary field with spatially varying autocorrelation. The script sweeps $\rho \in \{0.2, 0.5, 0.8\}$, $n \in \{400, 900, 1600\}$ and $\sigma_u \in \{0, 0.5, 1, 2\}$ for 20 replicates per cell and family, producing 2,880 SCCA fits with known ATT $= 0.5$. For each fit it records material bias (relative bias $> 25\%$) and whether candidate thresholds $t \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$ fire. Results are stored in `07_results/threshold_calibration_summary.csv`, `07_results/threshold_calibration_roc.csv`, and `07_results/threshold_calibration_roc_by_family.csv`.

Table 6 reports the pooled trade-off and Figure 6 the ROC. Pooling across families, the Youden's J optimum sits at $t = 0.05$ (TPR = 0.79, FPR = 0.03), while $t = 0.10$ keeps a low false-positive rate (0.02) and recovers 68% of materially biased fits. The 2026-06-30 rule set uses $t_{\text{Moran}} = 0.10$ as the default conservative threshold and treats 0.20 as a high-specificity sensitivity threshold.

The more important result is that this trade-off is *family-dependent* (Table 7). When the latent process is a distance-based kernel or non-stationary field, residual Moran's I is large under bias (mean 0.32 and 0.40) and the rule is highly sensitive even at $t = 0.20$ (TPR ≥ 0.99). When the latent process is a CAR field, residual Moran's I stays small under the same relative bias (mean 0.04), so the rule is weak: at $t = 0.10$ it recovers only 19% of biased CAR fits, and at $t = 0.20$ essentially none. SAR sits between these extremes (TPR 0.53 at $t = 0.10$). This is a direct external-validity limitation: the residual-Moran downgrade is a useful but not universal detector, and it is least sensitive precisely when the omitted spatial structure has weak simple-contiguity autocorrelation. We therefore present the rule as a calibrated warning boundary whose power depends on the unknown spatial process, and recommend pairing it with the neighbor-exposure and spatial-lag diagnostics rather than relying on residual Moran alone.

Table 6: Pooled calibration of the SCCA residual-Moran downgrade rule across four spatial-process families (2,880 fits). Positive class: relative bias > 25%. Predictor: $|I_{\text{residual}}| \geq t$ AND permutation $p \leq 0.05$.

Threshold t	TPR	FPR	Precision	Youden's J	Verdict
0.05	0.79	0.030	0.99	0.762	pooled Youden optimum
0.10	0.68	0.023	0.99	0.657	manuscript default; flags Chongqing
0.15	0.59	0.019	0.99	0.573	
0.20	0.57	0.018	0.99	0.556	high-specificity sensitivity
0.25	0.55	0.014	0.99	0.534	
0.30	0.50	0.008	0.99	0.494	too lenient

Table 7: Per-family behaviour of the residual-Moran rule. Mean residual Moran's I is averaged over all cells of that family; TPR is at the manuscript default $t = 0.10$. The rule is strong for kernel/non-stationary processes but weak for CAR, where the same relative bias leaves little simple-contiguity autocorrelation in the residual.

Latent-process family	Mean residual I	TPR @ 0.10	TPR @ 0.20	Family Youden-optimal t
CAR (conditional AR)	0.04	0.19	0.00	0.05
SAR (simultaneous AR)	0.10	0.53	0.30	0.05
Exponential kernel	0.32	0.99	0.99	0.20
Non-stationary	0.40	1.00	1.00	0.05

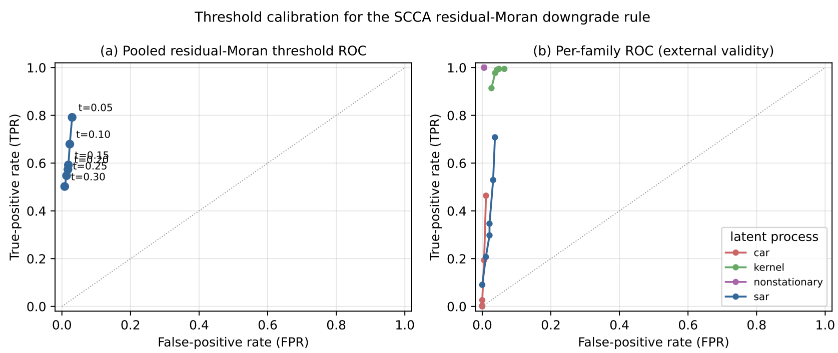

Figure 6: Threshold calibration for the SCCA residual-Moran downgrade rule. (a) Pooled ROC across candidate material thresholds. (b) Per-family ROC: the kernel and non-stationary families sit in the high-sensitivity corner, SAR is intermediate, and CAR is close to the chance-limited corner, showing that the rule's power depends on the unknown latent spatial process.

4.9 Grade-rule sensitivity for the two case studies

The county grade is stable, but the Chongqing grade is boundary-dependent. The sensitivity table varies only the material residual-Moran threshold. In Chongqing, the preferred pre-treatment specification's residual Moran's $I = 0.112$ ($p = 0.01$) triggers bounded support under the default 0.10 rule, but becomes a significant sub-threshold warning under 0.15 and 0.20. In the county case, residual Moran's $I = 0.313$ and significant neighboring exposure keep the grade bounded under all tested thresholds.

Table 8: Residual-spatial threshold sensitivity generated by the evidence-synthesis script. The exercise varies only the material residual Moran warning threshold and leaves all other downgrade rules unchanged.

Case	Residual Moran's I	Threshold	Grade	Residual status	Grade rules or flags
Chongqing pre-treatment	0.112	0.10	Bounded	material significant	`material_residual_moran`
Chongqing pre-treatment	0.112	0.15	Core	significant below material threshold	non-downgrade residual warning
Chongqing pre-treatment	0.112	0.20	Core	significant below material threshold	non-downgrade residual warning
County spatial notebook	0.313	0.10	Bounded	material significant	`material_residual_moran; significant_neighbor_exposure`
County spatial notebook	0.313	0.15	Bounded	material significant	`material_residual_moran; significant_neighbor_exposure`
County spatial notebook	0.313	0.20	Bounded	material significant	`material_residual_moran; significant_neighbor_exposure`

The county analysis also included response-shape threshold checks for the social-capital exposure. In the ArcGIS-trimmed sample, a knot at 13 gave an after-knot slope increase of 0.104 years per exposure unit (HC3 $p = 0.00075$; AIC improvement $= 17.46$), whereas knots at 10, 15, and 20 changed the magnitude and statistical strength. These checks support the county run as an operational sensitivity demonstration, not as independent evidence that the county exposure-response relation is causally identified.

4.10 GIS and notebook outputs

The county notebook run also tested SCCA as an operational GIS workflow. The shared core wrote joined tables, GeoPackage, GeoJSON, Shapefile, static maps, interactive maps, and QGIS `.qml` styles from the same `AnalysisRequest`. These outputs do not strengthen identification by themselves; they make assumptions, spatial residuals, target-exposure consequences, and neighborhood sensitivity visible in the environments where geographic analysts work.

5 Discussion

5.1 What SCCA solves

SCCA addresses a practical gap in geographic causal analysis: spatial variables are often added without showing whether they improved the adjustment design, and spatial residuals are treated as model diagnostics rather than boundaries on causal interpretation. SCCA makes the adjustment set inspectable: readers can see which context sources were tried, which variants passed balance, how many units remained in support, whether spatial checks were stable, and why the final claim was graded as core or bounded.

The Chongqing case illustrates the value of this discipline in the setting where the paper's main empirical claim is made. The most credible current estimate is not the best-balanced estimate. It is the estimate attached to a *pre-treatment* confounder set (coordinates, geometry, terrain), chosen for causal defensibility rather than for minimum SMD, with an explicitly audited variable-role table, an honestly reported statistically significant residual Moran's $I = 0.112$ ($p = 0.01$), an explicit change-of-support discount (cluster-robust and pixel-aggregated estimands), and a bounded evidence grade under the default rule. Adding the best-balanced Sentinel surfaces would attenuate the estimate from 0.303 to 0.244°C; SCCA's contribution here is not a single coefficient but the diagnosed record that shows why the pre-treatment coefficient, not the better-balanced one, is the defensible number to report.

The county notebook case shows the same logic for practical GIS use. A non-spatial summary would have looked stronger than warranted; adding spatial diagnostics kept the positive coefficient but downgraded support because residual autocorrelation and neighboring exposure remained.

5.2 Practical value for GIS workflows

SCCA converts a causal analysis into an inspectable spatial evidence package. The ArcGIS toolbox, QGIS provider, notebook workflow, and command-line interface share the same request object and output manifest, reducing silent divergence across notebooks, desktop joins, and publication tables. One run can produce estimates, diagnostics, joined GIS tables, spatial layers, maps, and QGIS symbology without rewriting the estimator or requiring ArcGIS to audit the result.

The ArcGIS-compatible county comparison gives a concrete example of this value. SCCA reproduced the main positive exposure-response direction and sample-trimming count of a commercial GIS analysis, then added explicit robustness, knot-sensitivity, and spatial-diagnostic outputs. This does not mean SCCA fully replaces the commercial desktop product or the substantive judgment required in GIS analysis. It means the same analysis can be inspected, rerun, and downgraded under declared rules in an open workflow.

GIS outputs are therefore audit objects, not causal evidence by themselves. They show where target-exposure changes, indirect-effect proxies, and residual spatial cautions occur, keeping the evidence boundary attached to the spatial objects used in analysis.

5.3 How SCCA can be extended

The current workflow can be strengthened without changing its core claim. Future versions should add sensitivity analysis for unmeasured spatial confounding, make variable-role declarations mandatory before adjustment, use spatial cross-fitting when high-dimensional nuisance models are introduced, and separate diagnostic SLX-style neighborhood adjustment from formal interference estimands [Chernozhukov et al., 2018, Forastiere et al., 2021, Papadogeorgou et al., 2022]. Scale also remains central: future applications should treat buffer radius, aggregation and neighborhood definitions as part of the adjustment design rather than as preprocessing details.

5.4 What SCCA does not solve

SCCA does not make observational spatial data equivalent to randomised experiments. It cannot adjust for confounders absent from all context sources or automatically solve interference. The SLX-style direct, indirect, and total summaries are sensitivity outputs, not identified network-interference estimands [Forastiere et al., 2021, Papadogeorgou et al., 2022]. This limit is concrete in the Chongqing case: heat generated by one high-rise cluster plausibly raises the LST of adjacent pixels, so the no-interference (SUTVA) assumption is unlikely to hold exactly. The consequence is that the building-level ATT conflates a unit's own effect with spillover from neighbours; the significant neighbor-exposure and spatial-lag terms are read here as a warning that this contamination is present, and the pixel-aggregated estimand (Section 4) partially absorbs local spillover by moving to the outcome's own resolution but does not identify a formal spillover effect. SCCA also does not remove MAUP or treatment-outcome scale mismatch [Openshaw, 1984, Fotheringham and Wong, 1991]. Context variables are audited by variants rather than selected automatically. Missing context values use within-variant median imputation with a missingness indicator (Section 3.9); spatially structured missingness should trigger a complete-case sensitivity rerun. The evidence-grade thresholds are machine-readable and predeclared, but only the residual-Moran rule is calibrated here, and that calibration shows it is process-dependent (weak under CAR-type structure); the boundary-mass, balance-correlation, and coefficient-change thresholds remain conservative manuscript defaults.

The Chongqing case illustrates these limits directly. Its residual Moran diagnostic ($I = 0.112$, $p = 0.01$ for the preferred pre-treatment set) is statistically significant and crosses the default calibrated threshold of 0.10. The correct interpretation is therefore a modest measured-context association with four stated boundaries: residual spatial structure that the SCCA-observed context did not absorb, change-of-support between the building-level treatment and the $\sim$1 km LST outcome (so the identified-scale estimand differs from the building-level one), possible post-treatment ambiguity that led us to exclude Sentinel surfaces from the preferred set, and sensitivity of the grade to the chosen material-Moran threshold. None of these is a defect of the workflow; they are properties of the data that SCCA exists to make visible.

6 Conclusions

We presented Spatial Context Causal Adjustment, an open diagnostic workflow for geographic observational causal studies. SCCA formalises spatial context as a candidate adjustment source and evaluates each adjustment variant through common support, balance, robustness, spatial diagnostics, variable-role audits, and reproducible evidence-grade rules. We motivated the residual-Moran downgrade rule as a diagnostic boundary and calibrated its default threshold against a 2,880-fit controlled design across four spatial-process families, which also revealed that the rule's power is process-dependent. Across synthetic benchmark stress tests, a Chongqing urban heat case, and a county social-capital notebook/GIS check, SCCA distinguished useful estimates from bounded and fragile designs.

The main empirical result is deliberately bounded. In Chongqing, the preferred pre-treatment specification produced a modest positive high-rise association with summer LST (ATT = 0.303°C) that is bounded under the default 0.10 residual-Moran rule; the residual Moran's $I = 0.112$ ($p = 0.01$) is statistically significant, and the estimate's magnitude is small relative to the thermal-retrieval and change-of-support uncertainties in building-to-$\sim$1 km LST extraction. The synthetic benchmark exposed estimator fragility under stress; a clean synthetic positive control was certified as core support, confirming the grade is discriminative; and the county notebook/GIS run showed practical ArcGIS-free reproducibility while demonstrating why a non-spatial strong result should be downgraded when residual spatial structure remains visible. The multi-family threshold calibration further shows that the residual-Moran rule is a useful but process-dependent detector – strong under kernel and non-stationary latent fields, weak under CAR fields – which is why SCCA pairs it with neighbor-exposure and spatial-lag diagnostics. Taken together, these findings support SCCA as a reproducible audit method for spatial causal adjustment. Its value is not that it removes the need for causal assumptions, but that it makes those assumptions, diagnostics, and claim boundaries visible in GIS-facing workflows.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The author thanks colleagues for early discussions of the SCCA workflow and the IJGIS reviewer pool for the structured feedback that motivated the threshold calibration experiment and variable-role audit in the present revision.

Funding

No external funding was used to support this work.

Data and Code Availability

The repository contains manuscript source, experiment scripts, generated result tables, diagnostic reports, synthetic generators, and redistributable public/example data. It supports exact reruns for the synthetic and county/example analyses and structural Chongqing reruns from approved local inputs. Raw Chongqing geospatial inputs and the building-level UHI sample are not publicly redistributed because they include precise building geometries or coordinates, floor attributes, and derived environmental variables. For review, the repository includes a non-sensitive aggregate Chongqing audit package with sample size, treatment threshold, caliper, matched counts, ATT, confidence interval, balance metrics, residual Moran diagnostics, and threshold-placebo summaries in `07_results/chongqing_reviewer_audit_package.csv` and `07_results/chongqing_reviewer_audit_package.json`. The public repository therefore does not claim full public reconstruction of the restricted Chongqing result.

The county social-capital case uses third-party Esri training/demo data rather than author-generated data. Its metadata credit Esri; the U.S. Census Bureau; NOAA/NOS/NGS; CDC WONDER/NCHS; and 2019 County Health Rankings/Living Atlas data from the University of Wisconsin Population Health Institute and the Robert Wood Johnson Foundation. Use is governed by the Esri Master License and restricted to training, demonstration, and educational purposes. Revised evidence synthesis, rule tables, variable-role audit, method comparison, and public result summaries are stored in `07_results/`.

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